A mod-11 modularity match over $\mathbb{Q}(\sqrt{5})$ from a $(1,11)$-polarized abelian surface

(working note)

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Abstract

We exhibit computational evidence for Serre's modularity conjecture over the real quadratic field $K = \mathbb{Q}(\sqrt{5})$. Starting from a point of the Gross–Popescu moduli of $(1,11)$-polarized abelian surfaces, we construct a genus-2 curve C over K whose Jacobian $A = \mathrm{Jac}(C)$ is geometrically simple ($\mathrm{End}_{\bar{K}} A = \mathbb{Z}$) with good reduction at both primes above 11. The mod-11 representation $A[11]$ splits over K as a sum of two irreducible 2-dimensional pieces ρ_1, ρ_2 with cyclotomic determinant — a splitting forced not by extra endomorphisms but by the polarization, since 11 is the polarization degree. Working entirely mod 11 inside a space of Hilbert modular forms computed via a definite quaternion algebra, we analyze whether ρ_1 and ρ_2 are realized by Hilbert modular eigenforms of parallel weight 2.

Correction (2026-06-29). An earlier version of this note claimed a match (modularity of ρ_1, ρ_2 up to a quadratic twist at 2, at level $\mathfrak{p} \cdot (2)$). *That claim was a false positive and is retracted.* The acceptance criterion used — nonvanishing of a "twist-robust" intersection V — is too weak, and the correct per-eigenform test fails (Section 5). The structural results on A and on $A[11] = \rho_1 \oplus \rho_2$ (Sections 2–3) are unaffected; the modularity of ρ_1, ρ_2 over $\mathbb{Q}(\sqrt{5})$ is, at present, *not* established by this computation. What we *can* do is derive, by elimination, the precise modularity statement that must hold (weight, character, level up to one bounded 2-adic exponent); see Section 6.

1 Introduction

Serre's modularity conjecture, a theorem over $\mathbb{Q}$ (Khare–Wintenberger), remains largely open over totally real fields. In its Hilbert-modular form it predicts that every odd, irreducible, continuous representation $\bar{\rho}\colon \mathrm{Gal}(\bar{K}/K) \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$ of the absolute Galois group of a totally real field K arises from a Hilbert modular eigenform. This note records a concrete instance over $K = \mathbb{Q}(\sqrt{5})$ with $\ell = 11$, obtained by the following pipeline:

$$(1,11) \text{ Gross–Popescu point} \longrightarrow \text{abelian surface } A/K \longrightarrow A[11] = \rho_1 \oplus \rho_2 \longrightarrow \text{Hilbert modular form.}$$

The point of the construction is that the abelian surface comes with a canonical mod-11 structure (the $(1,11)$-polarization), which is exactly what makes $A[11]$ reducible into 2-dimensional pieces and supplies interesting $\bar{\rho}$'s to test.

2 The abelian surface

Let $K = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$, of class number 1 and discriminant 5. The prime 11 splits in K.

The Gross–Popescu construction (`arXiv:math/9902017`) realizes a $(1,11)$-polarized abelian surface in $\mathbb{P}^{10}$ from a point of the sextic "odd 2-torsion" locus $D_2 \subset \mathbb{P}^4$. Inverting the associated theta map numerically yields a period matrix τ, hence (via its principally polarized partner) a genus-2 curve C/K. We work with the curve **Curve A**, with absolute Igusa–Siegel invariants (normalized $J_2 = 1$)

$$\left[1,\ \tfrac{243125\sqrt{5}-482787}{1459240},\ \tfrac{-54798934\sqrt{5}+127710391}{2229718720},\ \tfrac{182422196780\sqrt{5}-406668692089}{8517525510400},\ \tfrac{38134182372761\sqrt{5}-85270624049895}{65073894899456000}\right].$$

Proposition 1. *$A = \mathrm{Jac}(C)$ satisfies:*

(i) A is geometrically simple with $\mathrm{End}_{\bar{K}}(A) = \mathbb{Z}$ (no real multiplication);

(ii) A has good reduction at both primes of K above 11;

(iii) away from 11, the conductor of A has odd part $\mathfrak{p}$, a prime of K of norm 66179, at which A has multiplicative reduction (conductor exponent 1);

(iv) A also has (model-)bad reduction at the inert prime (2).

These are verified by computation: (i) via the geometric endomorphism algebra (CHIMP), (ii) and (iii) via the Igusa valuation criterion and conductor exponents.

3 The mod-11 representation

Let $\bar{\rho}\colon \mathrm{Gal}(\bar{K}/K) \to \mathrm{GSp}_4(\mathbb{F}_{11})$ be the representation on $A[11]$, and let χ_{11} denote the mod-11 cyclotomic character. We summarize the structure, all of which is verified over the 290 good primes P of K with $N(P) \leq 2000$ by reducing C and factoring the Frobenius characteristic polynomial $h_P(x) \in \mathbb{F}_{11}[x]$.

Proposition 2. *(a) For every such P, h_P factors over $\mathbb{F}_{11}$ as a product of two quadratics: its factor type is always $[1,1,2]$ or $[1,1,1,1]$, and never $[2,2]$, $[1,3]$, or $[4]$.*

(b) $\bar{\rho}$ has no 1-dimensional K-subquotient: there is no Galois character λ that is, at every prime, a root of the "linear part" of h_P (an exhaustive search over Hecke characters of K of order dividing 10 and every plausible conductor gives at best $90/132$ agreement).

(c) Consequently $A[11]$ is, up to semisimplification, a sum $\rho_1 \oplus \rho_2$ of two 2-dimensional representations, each with $\det \rho_i = \chi_{11}$ (hence odd). The splitting is supplied by the polarization: the kernel $K(\mathcal{L}) \cong (\mathbb{Z}/11)^2$ of the $(1,11)$-polarization is a K-rational 2-dimensional subspace of $A[11]$. The prime 11 is special precisely because it is the polarization degree; at other primes ℓ the image of $\mathrm{Jac}(C)[\ell]$ is the full $\mathrm{GSp}_4(\mathbb{F}_\ell)$.

(d) ρ_1, ρ_2 are irreducible over K, are not isomorphic, and are not quadratic twists of one another; their image is large (not dihedral/CM: the surface trace vanishes at only $\approx 10\%$ of primes).

Because $\det \rho_i = \chi_{11}$, the two traces are the roots of an explicit quadratic. Writing $h_P(x) = x^4 + Ax^3 + Bx^2 + Cx + D$ (so $D = N(P)^2$ and $C = N(P)\,A$), one has

$$\{\,\mathrm{tr}\,\rho_1(\mathrm{Frob}_P),\ \mathrm{tr}\,\rho_2(\mathrm{Frob}_P)\,\} = \text{roots of }\ y^2 + A\,y + \big(B - 2N(P)\big) \pmod{11}. \tag{1}$$

We call $\big(\mathrm{Tr}_P, \mathrm{Nm}_P\big) := \big(-A,\ B - 2N(P)\big) \bmod 11$ the *fingerprint* of $\{\rho_1, \rho_2\}$ at P. A sample:

$N(P)$	49	169	29	29	31	31	41	41
Tr_P	9	6	7	9	5	0	9	2
Nm_P	9	5	1	7	4	10	0	3

4 The modularity match

By good reduction at 11 (Proposition 1(ii)) each ρ_i is finite-flat at 11, so the predicted Serre weight is parallel weight 2, and the predicted level is prime to 11 and divides the conductor of A, i.e. $\mathfrak{p}\cdot(2)^a$ for some $a \geq 0$.

Let $S_2(\mathfrak{n})$ be the space of Hilbert cusp forms over K of parallel weight 2 and level $\mathfrak{n}$, computed via the definite quaternion algebra ramified at the two archimedean places, with its commuting Hecke operators T_P (Brandt matrices), which are integral. Reduce everything mod 11. If a Hilbert newform f has $\bar\rho_{f,\lambda} \cong \rho_i$ for some degree-one prime $\lambda \mid 11$ of its Hecke field, then its reduction is an eigenvector v in $S_2(\mathfrak{n}) \otimes \mathbb{F}_{11}$ with $T_P v = t_P\, v$ where t_P is a root of (1); that is, $\big(T_P^2 - \mathrm{Tr}_P\, T_P + \mathrm{Nm}_P\big)v = 0$.

Empirically the matching forms realize ρ_i only after a *quadratic twist* ε ramified at 2, so t_P is replaced by $\varepsilon(P)\, t_P = \pm t_P$. We therefore intersect the twist-robust kernels: set

$$V \;=\; \bigcap_P \Big[\; \ker\!\big(T_P^2 - \mathrm{Tr}_P\, T_P + \mathrm{Nm}_P\big) \;+\; \ker\!\big(T_P^2 + \mathrm{Tr}_P\, T_P + \mathrm{Nm}_P\big) \;\Big] \;\subseteq\; S_2(\mathfrak{n}) \otimes \mathbb{F}_{11}. \qquad (2)$$

Nonvanishing of V was the (flawed) criterion used in the retracted version; we now show it is too weak.

5 Refutation of the claimed match

The criterion is too weak. For each P, $\ker\big((T_P^2 - \mathrm{Tr}_P T_P + \mathrm{Nm}_P)(T_P^2 + \mathrm{Tr}_P T_P + \mathrm{Nm}_P)\big)$ is the kernel of a degree-4 polynomial in T_P; over $\mathbb{F}_{11}$ its dimension is roughly $\frac{4}{11}\dim S_2(\mathfrak{n})$. Intersecting a few such kernels can leave a nonzero V for reasons unrelated to any eigenform whose twisted eigenvalues match the fingerprint (1). One must test the eigenforms in V directly.

The dim-2 space does not realize $\rho_1 \oplus \rho_2$. At $\mathfrak{n} = \mathfrak{p}\cdot(2)$ one does find $\dim V = 2$, but restricting T_P to V gives a 2×2 matrix $C_P/\mathbb{F}_{11}$ (script `refine_serre3.m`) for which, at most primes,

$$\det C_P \neq \mathrm{Nm}_P, \qquad \mathrm{charpoly}(C_P) \notin \{q_P^+, q_P^-\},$$

indeed $\mathrm{charpoly}(C_P) \in \{q_P^+, q_P^-\}$ at only 2/25 test primes (both degenerate, $\mathrm{Nm}_P = 0$), and C_P usually has a *repeated* eigenvalue (e.g. $(y-7)^2$). So V is not semisimple and is not a sum of two genuine Hecke eigenforms realizing ρ_1, ρ_2.

No eigenform matches, even up to twist. The untwisted intersection $V^+ := \bigcap_P \ker(T_P^2 - \mathrm{Tr}_P T_P + \mathrm{Nm}_P)$ vanishes at level $\mathfrak{p}$ and at $\mathfrak{p}\cdot(2)$ (scripts `refine_serre4.m`, `refine_serre5.m`). The unique twist-robust survivor at level $\mathfrak{p}$ is a *single* eigenform f ($\dim V = 1$); its Hecke eigenvalue satisfies

$$a_P \in \{\pm t_1, \pm t_2\} \quad \text{at all 120/120 tested primes,}$$

a genuine structural fact (probability $\approx (4/11)^{106}$ by chance). Writing $\varepsilon(P) = +1$ if a_P is a root of q_P^+ and -1 if of q_P^-, the 94 unambiguous signs are *inconsistent with every quadratic ray-class character* of conductor supported in $\{2^{\leq 6}, 3, 5, 11, 66179, \infty\}$ (script `refine_charfit.m`); since K has narrow class number 1 there are no unramified quadratic characters either. So ε is not a quadratic Hecke character.

Proposition 3. *No Hilbert modular eigenform of parallel weight 2, trivial character, and level $\mathfrak{p}\cdot(2)^a$ realizes ρ_1 or ρ_2.*

Reason. Such a form has $\det \rho_f = \chi_{11} = \det \rho_i$, so $\rho_f \cong \rho_i \otimes \chi$ forces $\chi^2 = 1$: the only admissible twist is quadratic. A non-trivial nebentypus would change the determinant and so cannot occur. But a quadratic twist relating f to ρ_i is exactly what is excluded above (the sign ε is not a quadratic character), and the untwisted case is excluded by $V^+ = 0$. $\square$

The residual relation $a_P^2 \in \{t_1^2, t_2^2\}$ (the 120/120 fact) is the twist-invariant, projective (Sym^2) statement, with the square jumping between ρ_1 and ρ_2; it does *not* imply that either ρ_i is modular.

Remark 1. ***Status.*** *The mod-11 Serre-modularity match for Curve A is* not *established. The structural content (Propositions 1, 2) stands and is the reusable output of this work. Natural next steps: a genuine newform decomposition with eigensystems compared in the correct residue field (recall* 11 splits *in K, so a matching newform would have* 11 *split in its Hecke field and place both reductions in V^+, which is empty at the levels examined); searching other levels; and interpreting the a_P^2-relation. Reproducibility:* `mod11_rep.m`, `mod11_trnm.txt` *(target fingerprint), and the refutation scripts* `refine_serre[3,4,5].m`, `refine_charfit.m` *with raw signs in* `refine_serre5_signs.txt`.

6 The corrected modularity statement

Although we cannot exhibit the eigenform, the structural computations pin down what the correct statement must be. Assuming Serre's conjecture over $K = \mathbb{Q}(\sqrt{5})$ (each ρ_i is 2-dimensional, irreducible, and odd, so it applies), each ρ_i is modular, and its weight, character and level are forced up to a single bounded constant.

Theorem 1 (conditional on Serre's conjecture over $\mathbb{Q}(\sqrt{5})$)**.** *Each of ρ_1, ρ_2 is modular by a Hilbert modular eigenform over K of parallel weight $(2,2)$, trivial nebentypus, and level $\mathfrak{p} \cdot (2)^4$, realized untwisted ($\det = \chi_{11}$). Here $a_2(\rho_i) = 4$ is the conductor exponent at the inert prime* 2 *(Section 7).*

The parameters are forced as follows.

- **Determinant χ_{11}:** the two Frobenius eigenvalues of each piece multiply to $N(P)$ at all 132 primes of type $[1,1,2]$ (det-clean).
- **Trivial nebentypus:** $\det \rho_f = \psi\, \chi_{11}^{k_0-1}$, so weight 2 with $\det = \chi_{11}$ forces ψ trivial. Nebentypus χ_{11} would give $\det = \chi_{11}^2$; as χ_{11} has order 10 it has no square root among its powers, so no twist of ρ_i has that determinant.
- **Weight $(2,2)$:** A is *ordinary* at both primes above 11 ($L_{\mathfrak{l}}$ mod 11 has degree 2, i.e. two unit roots). Ordinary + good reduction $\Rightarrow$ peu-ramifiée; a subquotient of a finite-flat group scheme is finite-flat (as $e = 1 < 11 - 1$, by Raynaud); and $\det = \chi_{11}$ forces Hodge–Tate weights $\{0,1\}$. Hence Serre weight 2 at both primes.
- **Level $\mathfrak{p} \cdot (2)^a$:** pure 2-power levels $(2)^a$ have empty match, so ρ_i is ramified at $\mathfrak{p}$; and the untwisted match is empty at $\mathfrak{p} \cdot (2)^{0,1,2}$, so $a \geq 3$.

Remark 2. *Since the untwisted match is empty at 2-part ≤ 2 for both pieces, $a_2(\rho_1), a_2(\rho_2) \geq 3$, whence (Artin conductor is additive) $a_2(A) = a_2(\rho_1) + a_2(\rho_2) \geq 6$. Wild inertia at* 2 *acts through a 2-group of order ≤ 16 (the 2-Sylow of $\mathrm{GL}_2(\mathbb{F}_{11})$).*

7 The conductor at 2

We determine $c := a_{\mathfrak{p}_2}(A)$ numerically from the functional equation. Magma provides neither regular models nor minimisation for genus 2 over number fields, so we work with the degree-8 L-function $L(A/K, s) = L(W/\mathbb{Q}, s)$, $W = \mathrm{Res}_{K/\mathbb{Q}}A$, whose analytic conductor is

$$N(W) = |d_K|^4 \cdot N_{K/\mathbb{Q}}(\mathfrak{p} \cdot (2)^c) = 5^4 \cdot 66179 \cdot 4^c.$$

The right twist. The reconstruction `Genus2CurveFromIgusa` returns a curve A_{rec} that is *additive of potentially good reduction* at $499, 13711, 88301, 231481, \ldots$ (a spurious quadratic twist by the residual model content g; the Igusa criterion misses these because potential-good passes it). The correctly twisted curve $A = A_{\mathrm{rec}} \otimes \chi_g$, with χ_g the quadratic character of $K(\sqrt{g})$, is good at all those primes, so its conductor is exactly $\mathfrak{p} \cdot (2)^c$ and $L(A/K, s)$ is tractable. Its Euler factors are the χ_g-twist of those of A_{rec}: $L_P^A(T) = L_P^{A_{\mathrm{rec}}}(\chi_g(P)\,T)$.

Result. Building the Euler product to norm $3{\cdot}10^5$ (25 981 factors) and running `CheckFunctionalEquation` at precision 3 (`cfe_final2.m`):

c	6	7	**8**	9
|CFE|	0.10	0.29	**0.0020**	0.23

$c = 8$ closes the functional equation about two orders of magnitude more sharply than any neighbour, and is consistent with $a_2(A) \geq 6$. Hence $a_{\mathfrak{p}_2}(A) = 8$, and (symmetrically) $a_2(\rho_1) = a_2(\rho_2) = 4$, giving the level $\mathfrak{p} \cdot (2)^4$ of Theorem 1. This is a numerical (functional-equation) determination, not a rigorous proof.

8 An explicit form for ρ_1

The level-$\mathfrak{p}$ eigenform with $a_P^2 \in \{t_1^2, t_2^2\}$, dismissed above as coincidental, is in fact a genuine twist of ρ_1; it was missed only because the character search there omitted the primes $499, 13711, 88301, 231481$ where the reconstruction's spurious twist χ_g ramifies. Sweeping the quadratic characters χ_d ramified only at 2 and matching $A \otimes \chi_d$ by the untwisted filter, exactly one survives — χ_{-1}, the character of $K(i)/K$ — and already at level $\mathfrak{p}$ (verified over 80 primes).

Proposition 4. *$\rho_1 \cong \rho_g \otimes \chi$, where $\chi = \chi_g \cdot \chi_{-1}$ is the quadratic character of $K(\sqrt{-g})$ and g is an explicit Hilbert modular newform over K of weight $(2, 2)$, trivial nebentypus, and level $\mathfrak{p}$ (norm 66179): it is one of the four newform orbits of $S_2(\mathfrak{p})$, with Hecke field $E = \mathbb{Q}[x]/(x^5 + 8x^4 + 19x^3 + 4x^2 - 32x - 23)$, in which 11 is totally ramified.*

Since 11 is totally ramified in E, g has a single mod-11 reduction (residue field $\mathbb{F}_{11}$), which is $\rho_1 \otimes \chi$; consistently, $a_2(\rho_1) = 4 = 2 \cdot (\text{conductor of } \chi_{-1} \text{ at } 2)$, so the twist χ_{-1} undoes the wild 2-ramification and lowers the level from $\mathfrak{p} \cdot (2)^4$ to $\mathfrak{p}$.

Remark 3. *The two pieces behave differently at 2: for ρ_2 no quadratic twist ramified at 2 reaches level $\mathfrak{p} \cdot (2)^2$, i.e. $a_2(\rho_2 \otimes \chi_d) \geq 3$ for all such χ_d, so ρ_2 is twist-minimal (supercuspidal-like) at 2. Its form lives at level $\mathfrak{p} \cdot (2)^3$ (dimension ≈ 88000), where a single Hecke operator already exceeds available memory; ρ_2 is thus characterised but not yet exhibited.*